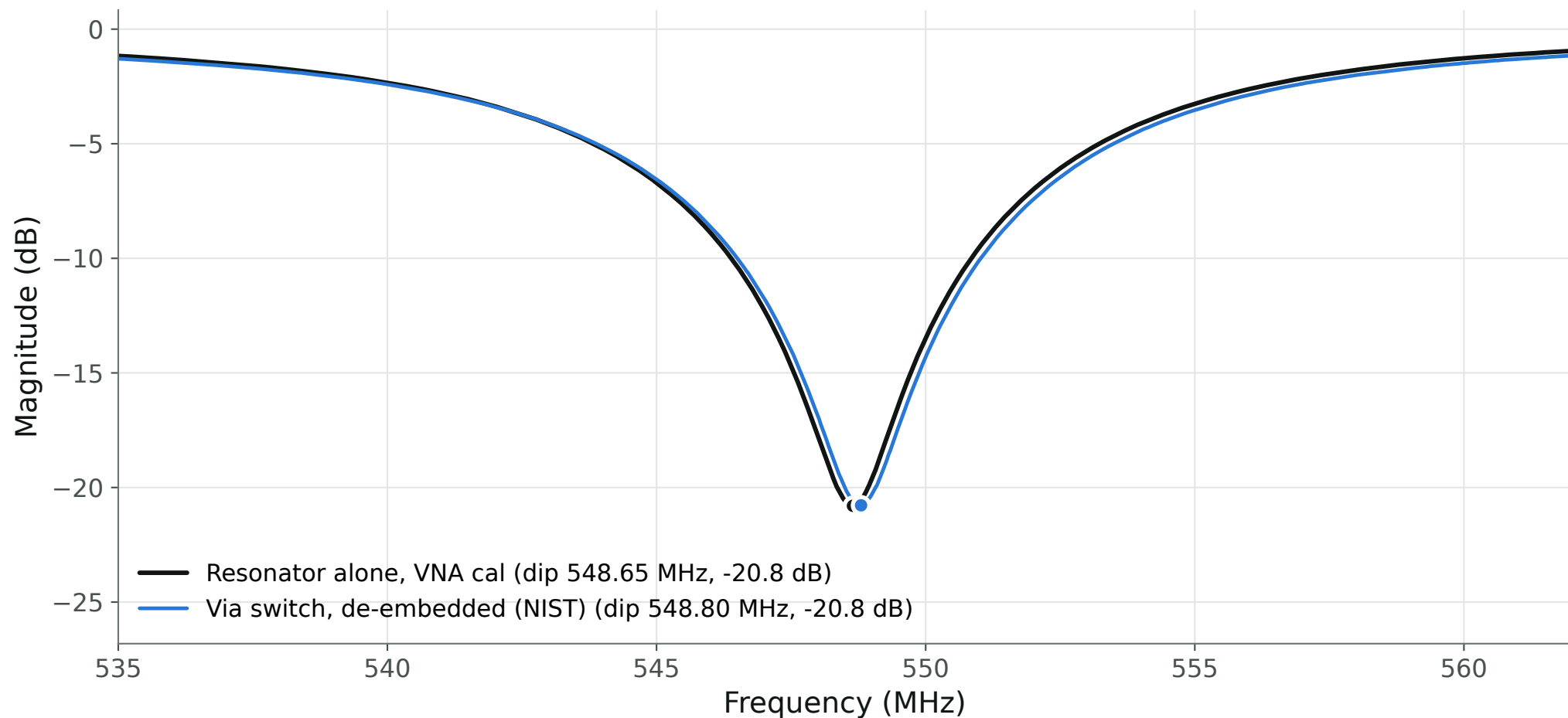


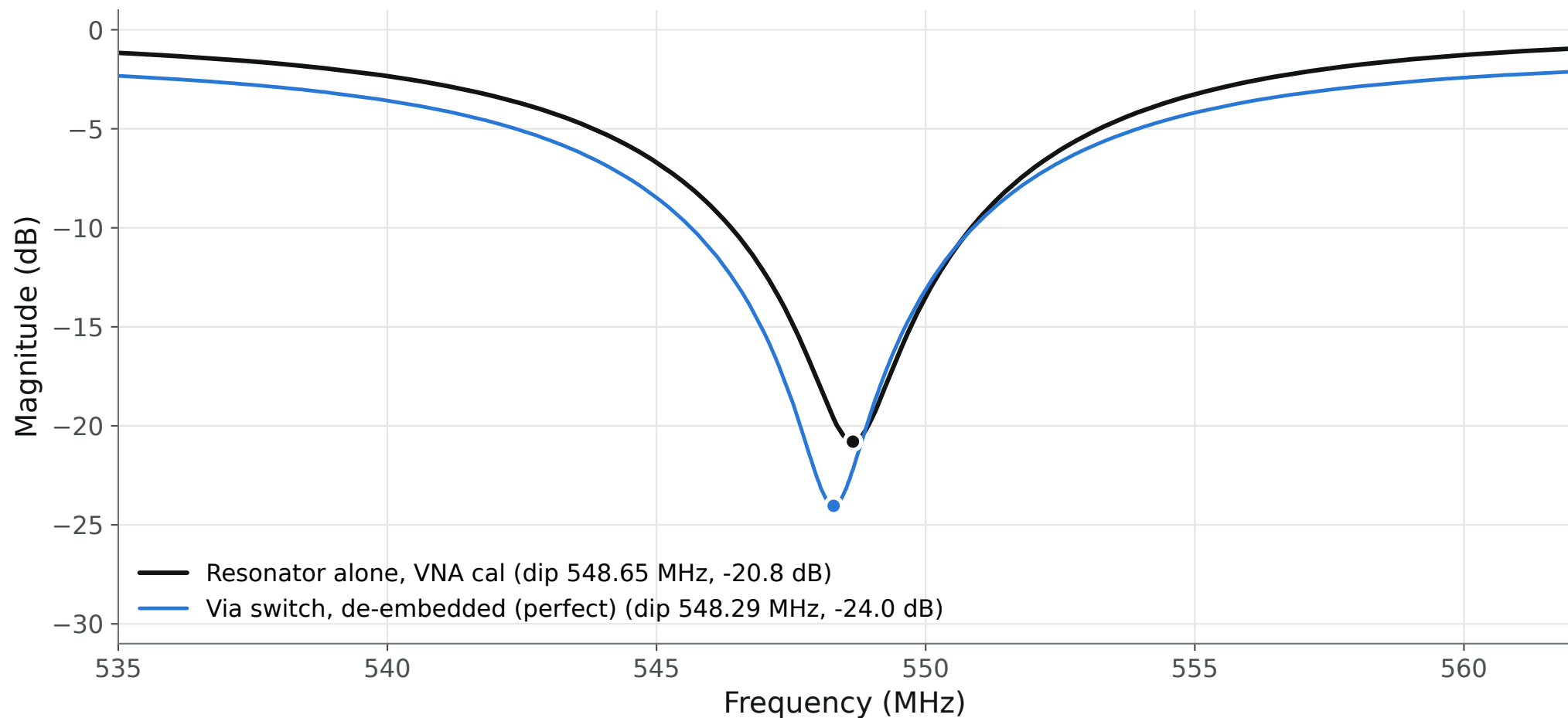
Resonator |S11|: VNA calibration vs. switch e-cal (NIST)

E-cal: NIST 295 K definitions (SN0031), plane at RF1 connector. Both repeats averaged.



Resonator |S11|: VNA calibration vs. switch e-cal (perfect)

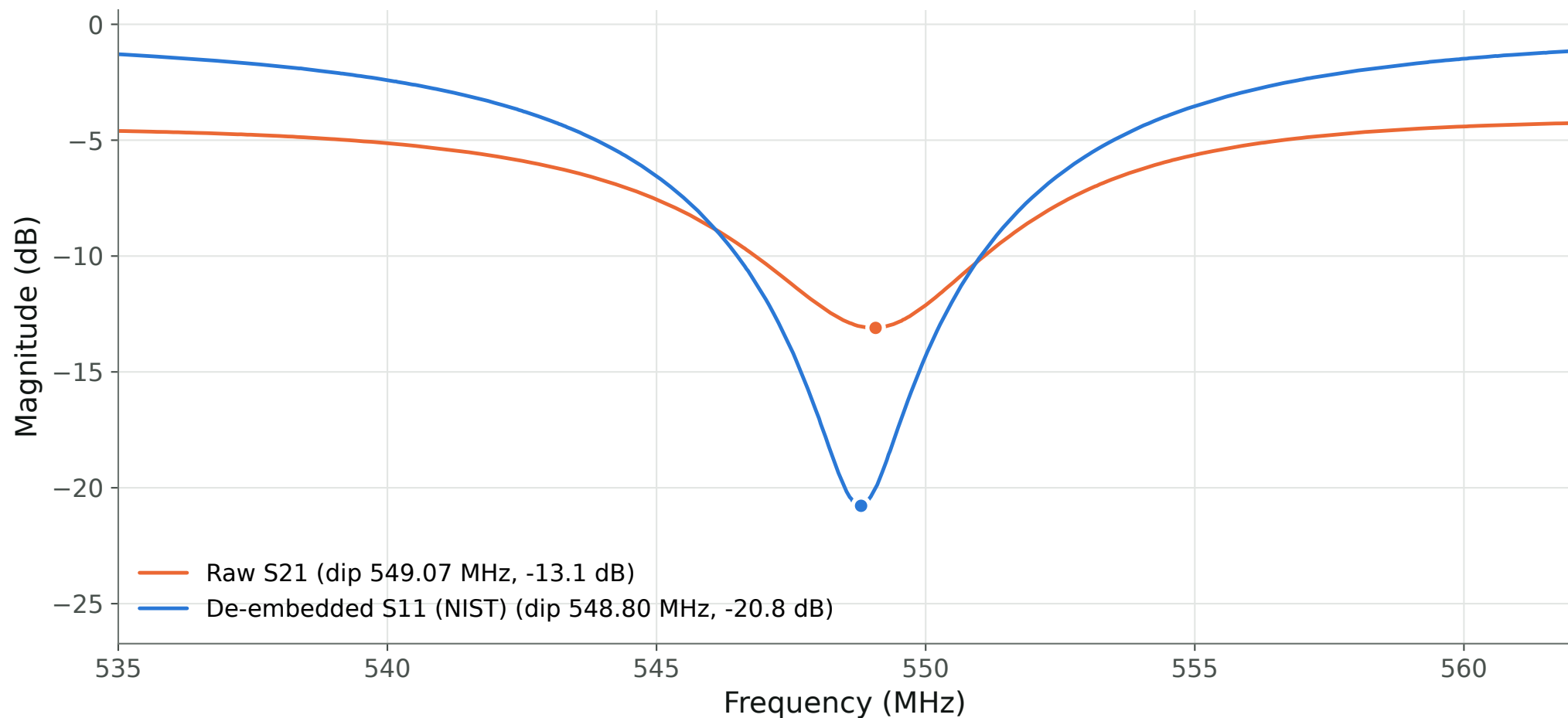
E-cal: perfect standards, plane inside the switch. Both repeats averaged.



Full setup: raw vs. de-embedded (NIST)

Raw = VNA port 1 -> circulator -> switch RF1 -> cable -> resonator -> back to port 2, VNA cal off.

De-embedded: NIST 295 K definitions (SN0031), plane at RF1 connector. Both repeats averaged.



Full setup: raw vs. de-embedded (perfect)

Raw = VNA port 1 -> circulator -> switch RF1 -> cable -> resonator -> back to port 2, VNA cal off.

De-embedded: perfect standards, plane inside the switch. Both repeats averaged.

